

Aditya Saini

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EDUCATION

Purdue University

West Lafayette, IN

B.S. Computer Science (Systems Programming)

Graduation: May 2028 (Expected)

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Mathematics, Computer Architecture, Programming in C, Object-Oriented Programming, Multivariate Calculus

EXPERIENCE

Embedded Engineer

Sep. 2025 – Present

Purdue Electric Racing

West Lafayette, IN

- Developed modular RTOS-based vehicle firmware in C, integrating drivers and performing in-circuit debugging
- Implemented runtime telemetry and data acquisition to profile and validate performance bottlenecks in production firmware
- Designed I2C and SPI HALs for STM32 platforms, standardizing high-frequency sensor communication
- Developed Nextion LCD UART drivers and optimized command encoding for reliable real-time dashboard updates

Chip Design Member (SoCET STARS)

June 2026 – August 2026

SoCET, Purdue VIP

West Lafayette, IN

- Designed combinational and sequential logic (decoders, registers, synchronizers, FSMs) in SystemVerilog RTL and validated each block with self-checking testbenches
- Progressed through the full digital ASIC design flow from RTL to GDSII with OpenLane, applying FPGA synthesis, timing analysis, and MMIO/driver/accelerator integration
- Wrote RISC-V assembly for the AFTx07 ISA, simulated gate-level circuits in LTSPICE, and managed RTL builds with Linux, Makefiles, and Git on the asicfab server

Robotic Safety Research Assistant

January 2026 – Present

PurSec, Purdue Computer Science

West Lafayette, IN

- Researched jailbreak vulnerabilities in multimodal-LLM-controlled robots, uncovering multi-step attacks that chain individually-safe actions to defeat state-of-the-art safety models
- Built containerized (Docker) Python pipelines running GPU-accelerated AI2-THOR simulation and model evaluation across 10+ scenarios on headless Linux cloud instances (AWS EC2), automating reproducible benchmarking at scale
- Aggregated per-step model verdicts into structured metrics and visualized results with pandas/Matplotlib to analyze failure modes across the ML training, evaluation, and inference lifecycle

PROJECTS

Natural Language Shell | C, Python, OpenAI API, Git

June 2025 – January 2026

- Created a lightweight command-line shell in C supporting natural language commands with tokenizer, intent parsing, and bash mapping; integrated OpenAI API for complex commands/queries through OpenRouter.

FRC Team 2554 | Java, OpenCV, Jetson Orin Nano, CUDA

September 2021 – August 2025

- Led the team to its best-ever season as Build President, bringing us to District Championships after 18+ years.
- Built a CUDA-accelerated OpenCV vision pipeline on a Jetson Orin Nano to detect AprilTags in real time and tuned PID/FF control, improving autonomous positioning accuracy by 66%.

Tava | React Native, Expo, TypeScript, Firebase, Mapbox, OpenAI

September 2025

- Built a cross-platform social app connecting student builders via an AI-powered search and Tinder-style swipe matching, with a node-edge network graph and interactive Mapbox map backed by Firebase auth and data.

TECHNICAL SKILLS

Languages: Python, C, C++, SystemVerilog, Java, SQL, JavaScript, HTML/CSS, R

ML & GPU: PyTorch, CUDA, OpenCV, NumPy, pandas, Matplotlib, model training/evaluation/inference

Systems & Tools: Linux, Git, Docker, containerization, AWS EC2 (GPU/HPC), Makefiles, RTOS, benchmarking

Embedded & Hardware: STM32, ESP32, SPI, I2C, UART, CAN, TCP/IP, FPGA, SystemVerilog RTL, LTSPICE